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LU - Factorization:

An LU-factorization of a given square matrix A is of the form

$$A = LU$$

where L is lower triangular and U is upper triangular for exp.

$$A = \begin{bmatrix} 2 & 3 \\ 8 & 5 \end{bmatrix} = LU = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 0 & -7 \end{bmatrix}$$

The crucial idea now is that L & U can be composed directly without solving simultaneous eq. (without using Gauss elimination)

We can solve $AX = B$ in two steps.

(i) $Ly = b$

(ii) $Ux = y$

Expt solve by LU Factorization.

$$x_1 + 2x_2 + 5x_3 = -11$$

$$x_2 = 1$$

$$2x_1 + 9x_2 + 11x_3 = -20$$

Sol

$$A = \begin{bmatrix} 1 & 2 & 5 \\ 0 & 1 & 0 \\ 2 & 9 & 11 \end{bmatrix} = LU = \begin{bmatrix} 1 & 0 & 0 \\ L_{21} & 1 & 0 \\ L_{31} & L_{32} & 1 \end{bmatrix} \begin{bmatrix} U_{11} & U_{12} & U_{13} \\ 0 & U_{22} & U_{23} \\ 0 & 0 & U_{33} \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 2 & 5 \\ 0 & 1 & 0 \\ 2 & 9 & 11 \end{bmatrix} = \begin{bmatrix} U_{11} & U_{12} & U_{13} \\ L_{21}U_{11} & L_{21}U_{12}+U_{22} & L_{21}U_{13}+U_{23} \\ L_{31}U_{11} & L_{31}U_{12}+L_{32}U_{22} & L_{31}U_{13}+L_{32}U_{23}+U_{33} \end{bmatrix}$$

$$\boxed{U_{11} = 1}, \quad \boxed{U_{12} = 2}, \quad \boxed{U_{13} = 5}$$

$$L_{21}U_{11} = 0, \quad L_{21}U_{12} + U_{22} = 1, \quad L_{21}U_{13} + U_{23} = 0$$

$$\boxed{L_{21} = 0}, \quad 0 + U_{22} = 1, \quad 0 + U_{23} = 0$$

$$\boxed{U_{22} = 1}, \quad \boxed{U_{23} = 0}$$

$$L_{31}U_{11} = 2, \quad L_{31}U_{12} + L_{32}U_{22} = 9, \quad L_{31}U_{13} + L_{32}U_{23} + U_{33} = 11$$

$$\boxed{L_{31} = 2}, \quad (2)(2) + L_{32}(1) = 9, \quad (2)(5) + (5)(0) + U_{33} = 11$$

$$\boxed{L_{32} = 5}, \quad \boxed{U_{33} = 1}$$

put all the values in eq (1) (2) (3)

$$A = \begin{bmatrix} 1 & 2 & 5 \\ 0 & 1 & 0 \\ 2 & 9 & 11 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 5 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 5 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Apply First step $Ly = b$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 5 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} -11 \\ 1 \\ -20 \end{bmatrix} \quad -3(-5/2)$$

By read Back

$$\boxed{y_1 = -11} \quad \text{--- (1)}$$

$$\boxed{y_2 = 1} \quad \text{--- (2)}$$

$$2y_1 + 5y_2 + y_3 = -20 \quad \text{--- (3)}$$

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$$\textcircled{3} \Rightarrow 2(-11) + 5(1) + y_3 = -20$$
$$\boxed{y_3 = -3}$$

Apply 2nd step $Ux = y$

$$\begin{bmatrix} 1 & 2 & 5 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -11 \\ 1 \\ -3 \end{bmatrix}$$

By read Back

$$x_1 + 2x_2 + 5x_3 = -11 \quad \text{---} \textcircled{4}$$

$$x_2 = 1 \quad \text{---} \textcircled{5}$$

$$x_3 = -3 \quad \text{---} \textcircled{6}$$

$$\textcircled{4} \Rightarrow x_1 + 2(1) + 5(-3) = -11$$

$$\boxed{x_1 = 2}$$

$$\text{sol is } \boxed{x_1 = 2 \mid x_2 = 1 \mid x_3 = -3}$$

check

$$x_1 + 2x_2 + 5x_3 = -11$$

$$2 + 2(1) + 5(-3) = -11$$

$$2 + 2 - 15 = -11$$

$$\boxed{-11 = -11} \text{ verify.}$$

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Ex 12 solve by LU-Factorization.

$$5x_1 + 4x_2 + x_3 = 3.4$$

$$x_1 = 0.2$$

$$10x_1 + 9x_2 + 4x_3 = 8.8$$

$$x_2 = 0.4$$

$$10x_1 + 13x_2 + 15x_3 = 19.2$$

$$x_3 = 0.8$$

Iteration Method.

When a no. of ~~system~~ ^{Eq^s} in a system is too large or when most of the element in a matrix are zero. Iterative method are particularly suitable.

Advantages

- * They ~~are~~ have less round off error than elimination method.
- * They are self-correcting if an error is made.
- * They are quicker & easier.

We consider only two classical iterative Method.

1) Jacobi's Method.

2) Gauss Seidal Method.

Let $Ax = B$

$$x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + x_2 + a_{23}x_3 + \dots + a_{2n}x_n = b_2$$

$$a_{31}x_1 + a_{32}x_2 + x_3 + \dots + a_{3n}x_n = b_3$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + x_n = b_m$$

Eq (2) can be written as.